Riya Basak

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Profile

My research focuses on self-supervised, structured world models for general intelligence: learning abstract representations and transferable causal dynamics for compositional generalisation, prediction, long-horizon planning, and embodied action in the physical world. I aim to translate these foundations into practical safety- and security-critical systems where reliable world understanding, prediction, and decision-making are essential.

Technical Foundations

Programming & Research Computing: Python, PyTorch, NumPy, Linux, Git/GitHub, Bash, LaTeX

Mathematical Foundations: Linear Algebra, Probability, Statistics, Calculus, Optimisation

Representation & World-Model Learning: Self-Supervised Learning, Structured Representation Learning, World Models, Compositional Generalisation, Causal Representation Learning

Planning & Sequential Decision-Making: Model-Based Prediction and Planning, Reinforcement Learning, Offline Reinforcement Learning, Safe Reinforcement Learning, Constrained Policy Optimisation

Embodied & Reliable AI: Embodied Agents, Action-Conditioned Prediction, Uncertainty-Aware Decision-Making, Counterfactual Safety Analysis

Research Methodology: Controlled Baselines and Ablations, Out-of-Distribution Evaluation, Predictive-Uncertainty Analysis, Failure-Mode Analysis, Reproducible Experimentation

Education

University of Hertfordshire — **Sep 2023 – May 2026**

BSc (Hons) Computer Science with Artificial Intelligence — First Class Honours — *Hatfield, UK*

GPA: Level 4: **4.34/4.50** | Level 5: **4.34/4.50** | Level 6: **4.44/4.50** | **Cumulative: 4.38/4.50**

U.S. Latin-Honours Equivalent: *Summa Cum Laude*

Final-Year Departmental Standing: Rank 1 (Top 1%) in Computer Science | **Rank 1 (Top 1%)** in Artificial Intelligence

Honours & Awards

University Graduation Prize — Rank 1 at Level 6 — **2026**

Highest Achievement in BSc (Hons) Computer Science (Artificial Intelligence) — *University of Hertfordshire*

Awarded by the Board of Examiners for the **highest achievement at Level 6** in the BSc (Hons) Computer Science (Artificial Intelligence) programme.

Experience

Mercor — **Sep 2025 – Nov 2025**

Artificial Intelligence Researcher — Mathematical Reasoning

- Investigated failure modes in AI mathematical reasoning by formulating Olympiad-level problems as explicit assumptions, constraints, intermediate claims and proof obligations.
- Derived and evaluated solution paths using constructive reasoning, case analysis, edge-case testing and rigorous assumption checking, then assessed AI-generated proofs for validity, completeness and efficiency.
- Converted recurrent errors—including unsupported implications, omitted cases and circular reasoning—into structured evaluation criteria for more precise analysis of mathematical reasoning failures.

Cubble — **Aug 2025 – Sep 2025**

Full Stack AI Intern — ML Systems Research & Evaluation

- Investigated how deployed AI failures could be inspected systematically rather than through ad-hoc debugging; translated the proposed evaluation methodology into model-review, safety-flag, logging and failure-analysis tools.
- Integrated frontend inspection with backend AI services and structured testing to trace prediction errors, recurring failure modes and application-layer faults reproducibly.
- Improved the research-to-deployment workflow by making model behaviour and failure causes directly inspectable, with clearer experimental documentation and validation.

Cubble **Jul 2025 – Aug 2025**
Software Engineer Intern — Machine Learning Research

- Investigated which learned visual representations generalised most reliably for safe/unsafe user-uploaded image classification under limited task-specific data; compared ResNet50 and CLIP-based approaches.
- Independently implemented transfer-learning pipelines and evaluated them through controlled validation, optimisation, quantitative error analysis and overfitting analysis to refine model design.
- Achieved **97.5%** accuracy on unseen images with stable validation behaviour; the AI team supervisor recognised my work as the top-performing implementation of the proposed working-model design.

Projects

Mitigating Shortcut Learning in Brain Tumour MRI Classification **Oct 2025 – Mar 2026**
*Dissertation Project **[PREPRINT] [CODE]*** *Shortcut Learning, Medical Imaging, Explainable AI*

- Proposed Pathology-Focused Disentanglement and Guided Semantic Token Evolution (PFD–GSTE), with variants A and B, for shortcut-learning mitigation without segmentation masks.
- Designed a ResNet50V2–RViT (CNN–Transformer) hybrid for four-class brain-tumour MRI classification; my literature review found no prior peer-reviewed study using this exact combination for the proposed shortcut learning mitigation experiment.
- Trained fixed-split PyTorch models with leakage-aware preprocessing and ablations; all models reached **98.52–99.22%** test accuracy and **98.49–99.20%** macro-F1.
- XAI analysis showed the main result: PFD–GSTE-B gave stronger tumour-region attribution than PFD–GSTE-A, with tumour-focused evidence in about **90%** versus **70%** of reviewed test cases.
- PFD–GSTE-B mitigated shortcut reliance more reliably on the tested benchmark; PFD–GSTE-A was weaker despite high classification accuracy.
- Also performed a small qualitative OOD demonstration on five external T1ce meningioma MRI images, kept separate from the formal benchmark evaluation and project submission.

Admissibility-Aware Offline Actor-Critic Learning for Safer ICU-Sepsis Treatment Decisions **Mar 2026 – Apr 2026**
*Proposed Coursework **[CODE]*** *Safe Offline Deep RL*

- Proposed Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), an offline actor-critic method for safer treatment-policy learning in the ICU-Sepsis benchmark.
- Built, to my literature-review findings, a new combination for ICU-Sepsis: hard admissibility masking, twin critics, conservative critic regularisation, expert-policy regularisation, smoothness shaping, and Lagrangian cost control.
- Compared LAADAN-AC with Behaviour Cloning, CQL-regularised fitted-Q, and Vanilla Offline Actor-Critic under the same dataset, seeds, horizon, and evaluation metrics.
- LAADAN-AC achieved the strongest safe policy: **0.7931** ± **0.0007** survival/return, **0.0000** inadmissibility, and the highest expert-action match at **0.9525**.
- LAADAN-AC gave the best return–safety trade-off, improving over BC and CQL while avoiding the unsafe behaviour of unconstrained actor-critic learning.

LAADAN-AC: Beyond Survival in Admissible Offline Treatment-Policy Learning **Apr 2026 – Jun 2026**
*Accepted ICaTAS 2026 **[CODE]*** *Safe Offline Deep RL, Cross-Domain Check*

- Extended LAADAN-AC into a reproducible research manuscript and codebase with component ablations, Lagrangian frontiers, safety-failure analysis, and a cross-domain portability check.

From World Models to Embodied Social Robots **May 2026 – Present**
Research Project — in Preparation for ICLR 2027 *Embodied AI, Social Robotics, World Models*

- Building a LeWM- and DreamerV3-based world-model stack for embodied agents that learn from predicted physical/social futures, using ARC Prize 2026 / ARC-AGI-3 as an initial abstract testbed before embodied AI and social robotics.
- Propose KCON: Kinetic Counterfactual Oversight Nexus, a compact executive layer between imagined futures and action choice, adding action-effect normlet memory, minimal intervention, uncertainty/novelty handling, and failure-aware selection.

Same Rules, New Worlds **Jul 2026 – Present**
Research Project (Ongoing) — in Preparation for CVPR 2027 *World Models, Compositional Generalisation*

- Investigating why world models that predict well in-distribution often fail to transfer learned dynamics across changes in context, appearance, embodiment and long-horizon interaction.
- Developing a mechanism-transport world model that separates persistent state from transferable dynamics, enforcing consistency across nuisance shifts while remaining sensitive to genuine changes in the underlying causal mechanism.
- Evaluating compositional and interventional generalisation under matched appearance–dynamics shifts, with controlled baselines, long-horizon tests and ablations before reporting quantitative claims.